# Final Project Report: Feature Extraction and Price Prediction for Mobile Phones

**1. Project Overview & Scenario**

- **Context**: Modern smartphone pricing is heavily segmented across hardware specifications and brand positioning.
- **Objective**: Identify the primary structural features driving handset prices, perform feature extraction and selection, and construct machine learning models capable of accurately predicting retail prices.
- **Dataset**: 541 records across 12 raw features including Model, Color, Memory (ROM), RAM, Battery, Rear Camera, Front Camera, AI Lens, Height, Processor, and Price.

**2. Data Cleaning & Feature Engineering**

1. **Outlier Correction**: Identified and rectified typographical errors in device dimensions (e.g., MOTOROLA G62 5G listed at 41.94 cm, imputed to true specification of 16.64 cm).
2. **Camera Megapixel Extraction**: Applied regular expressions to extract numeric megapixel integers (Rear_Camera_MP, Front_Camera_MP) from raw string values (e.g., '50MP').
3. **Brand Extraction & Tiering**: Extracted core brand identifiers from phone model names and grouped rare brands to prevent one-hot sparsity.
4. **Network Capability (Is_5G)**: Engineered a binary flag based on model and SoC designations.
5. **Processor Family Clustered**: Mapped 120+ unique chipset descriptions into primary architecture families (Snapdragon, MediaTek, Unisoc, Exynos, Google, Apple).

**3. Model Performance Benchmark**

Models were trained on an 80/20 train-test split using a Scikit-Learn ColumnTransformer (StandardScaler for numerical features, OneHotEncoder for categoricals):

| Model Algorithm | Test R2 Score | Mean Absolute Error (MAE) | Root Mean Squared Error (RMSE) |
|---|---|---|---|
| **Linear Regression** | 0.750 | ₹2,951 | ₹5,870 |
| **Ridge Regression** | 0.722 | ₹3,134 | ₹6,189 |
| **Decision Tree** | 0.858 | ₹2,069 | ₹4,418 |
| **Random Forest** | 0.858 | ₹1,601 | ₹4,422 |
| **Gradient Boosting (Champion)** | **0.882** | **₹1,280** | **₹4,038** |

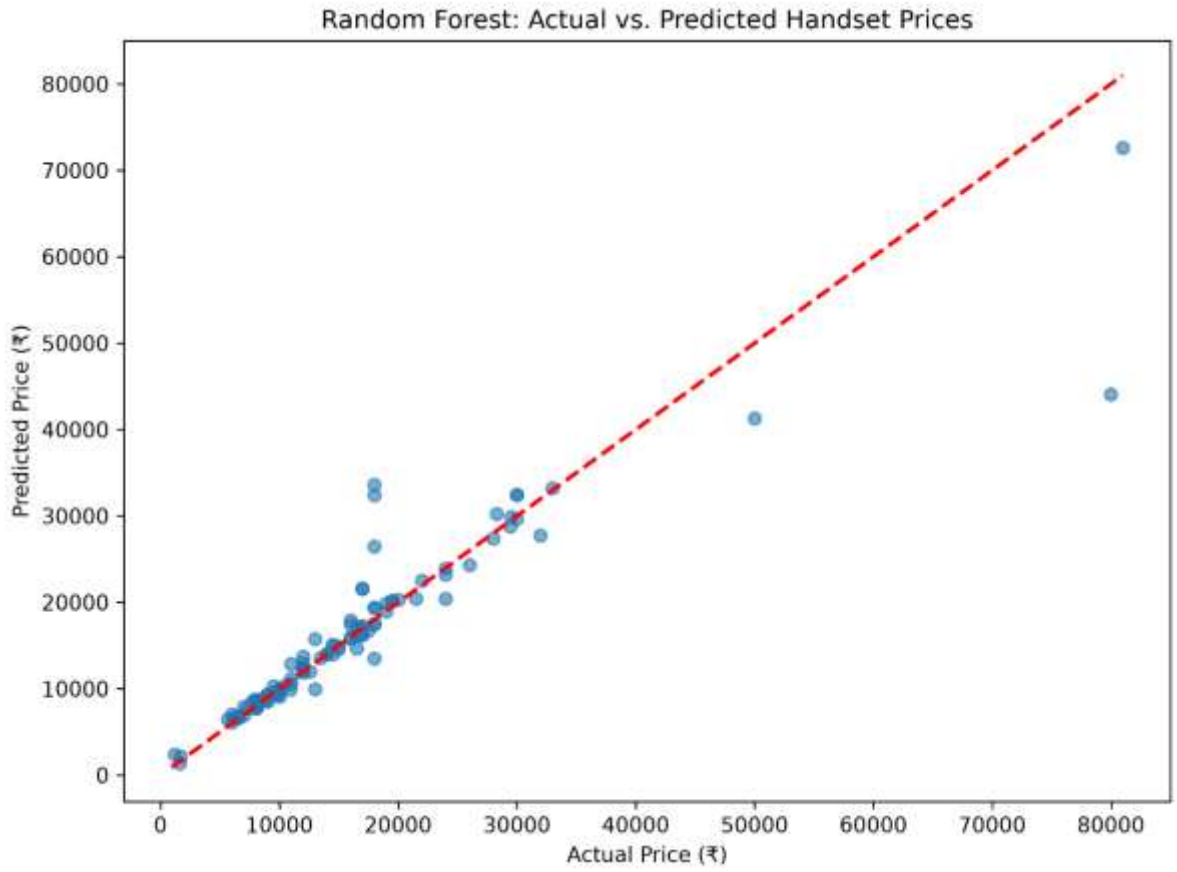
**Champion Model**: **Gradient Boosting Regressor** achieved the highest explained variance ($R^2 = 0.882$) and the lowest average deviation (MAE $= ₹1{,}280$), effectively capturing non-linear pricing jumps across spec thresholds.

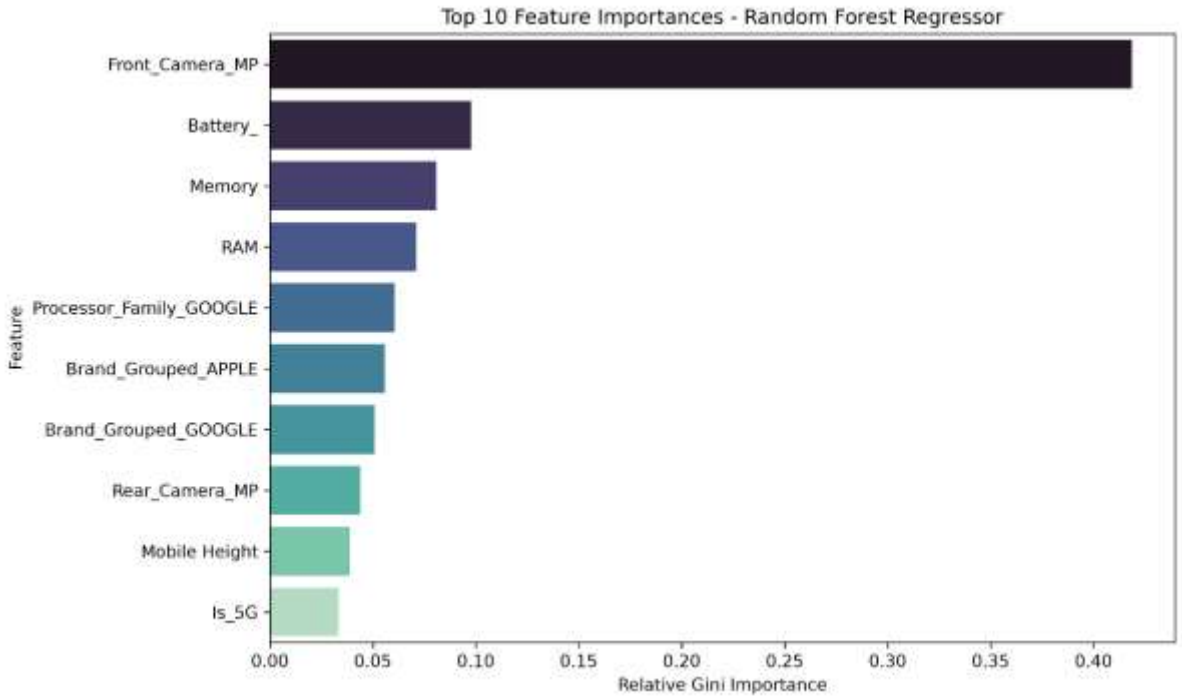
**4. Feature Importance Analysis**

From the Random Forest ensemble, the top 10 most influential predictors of smartphone price are:

1. **Front Camera MP (41.86%)**: The single highest predictor. High-end selfie sensors (32MP, 50MP) are almost exclusively reserved for premium devices, while budget models share commodity 50MP rear sensors.
2. **Battery Capacity (9.77%)**: Exhibits an inverse tier relationship—entry/mid-range devices prioritize high mAh capacity (5,000–6,000 mAh), whereas ultra-flagships balance smaller cells with fast charging and slim designs.
3. **Internal Storage / Memory (8.08%)**: Direct linear driver for SKU-level pricing.
4. **System RAM (7.11%)**: Delineates budget tiers (2GB–4GB) from performance handsets (6GB–8GB+).
5. **Google Processor / Tensor (6.06%)**: Premium SoC premium associated with Pixel flagship devices.
6. **Apple Brand Equity (5.58%)**: Substantial standalone pricing premium independent of raw spec numbers.
7. **Google Brand Equity (5.09%)**: Substantial pricing anchor in the mid-to-high tier.
8. **Rear Camera MP (4.39%)**: Influential, but secondary to front camera due to sensor commoditization.

9. **Mobile Height (3.88%)**: Correlates with display form factor and panel class.
10. **5G Capability (3.34%)**: 5G handsets maintain a distinct price premium over legacy 4G SKUs.






**5. Strategic Recommendations for the Organization**

- **Front Camera Optimization Over Rear Gimmicks**: Allocate bill-of-materials (BOM) budget toward high-grade front sensors (16MP–32MP) rather than adding low-cost auxiliary depth or AI lenses, as front camera quality directly anchors higher consumer price perception.
- **Monetize Storage Steps**: Price deltas between 128GB and 256GB configurations should be maintained at a premium margin, as internal storage remains one of the top direct purchase justifications.
- **Battery Capacity Strategy**: Do not attempt to charge a premium purely for battery capacity above 5,000 mAh; consumers in higher-price bands prioritize chassis thinness and charging speed over raw milliamp-hour ratings.
- **Tiered 5G Cutoff**: Position all non-5G inventory aggressively beneath the ₹12,000 price point, as devices above this threshold are expected to feature 5G connectivity as baseline criteria.